

ORIGINAL ARTICLE

Critical Audit Matters and SEC Filing Review

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ABSTRACT

This study explores the alignment between auditors' critical audit matters (CAMs) and the issues raised in SEC comment letters. CAMs highlight areas involving significant auditor judgement, whereas SEC comment letters address concerns related to compliance and the quality of financial disclosure. When deficiencies are identified, the SEC issues comment letters requesting clarification or additional information, a resolution process that reflects investor priorities. We find that SEC comment letters are more likely to reference the same financial statement note numbers referenced in CAMs. Additionally, we document strong associations between the accounting topics addressed in SEC comment letters and those disclosed in CAM reports. Furthermore, firms exhibiting higher uncertainty sentiment in CAMs are more likely to receive SEC comment letters on related issues. Our results support the informational validity and credibility of CAMs, suggesting that these reports highlight salient financial reporting issues that are relevant to investors.

JEL Classification: M41, M42, G38, M48

1 | Introduction

Firm governance, external audits and Securities and Exchange Commission (SEC) filing review collectively function as a multilayered monitoring mechanism designed to protect investors and ensure high-quality financial reporting. Each mechanism serves a distinct yet interrelated role: corporate governance structures guide the preparation of 10-K filings, auditors provide independent assurance over these filings, and the SEC reviews postaudit filings to ensure compliance, transparency and completeness (SEC 2019). When multiple layers of monitoring focus on similar areas of financial reporting risk, their combined monitoring can constrain misreporting (Fang et al. 2024). Bushman and Smith (2001) emphasize the importance of understanding the interplay among these governance mechanisms to enhance financial reporting quality, whereas DeFond and Zhang (2014) call for further research on how reporting regulations and enforcement mechanisms interact. We extend this literature by examining the alignment between issues identified in external

audits—specifically, critical audit matters (CAMs)—and those raised in SEC comment letters. Evidence of such alignment supports the informational validity and credibility of CAM reports, suggesting that they highlight salient financial reporting issues that are relevant to investors.

Auditors play a pivotal role by issuing independent audit reports that include disclosures of CAMs. In 2017, the Public Company Accounting Oversight Board (PCAOB) adopted Auditing Standard 3101 (AS 3101), which requires auditors to disclose CAMs that (1) relate to material accounts and disclosures and (2) involve challenging, subjective or complex auditor judgement in the auditor's report. The CAM reports, along with the auditor's opinion, are included in the firm's annual 10-K filing. The objective of CAM reporting is to reduce the information gap and increase transparency between investors and auditors, offering investors valuable insights into key risks and uncertainties in financial reporting (PCAOB 2024).¹ DeFond and Zhang (2014) suggest that the CAM reporting requirement marks a significant

departure from the boilerplate language that has traditionally characterized the auditor's report.

However, the effectiveness of CAM reporting in enhancing financial transparency and investor protection remains debated. Proponents argue that CAMs identify areas of significant reporting risk and auditor judgement. For example, Jahan and Karim (2024) find that firms disclosing goodwill-related CAMs are more likely to recognize goodwill impairment losses. On the other hand, critics raise concerns about boilerplate reporting, information overload and unintended consequences for audit quality and financial reporting (e.g., Chan and Liu 2023; Lennox et al. 2023; Coram and Wang 2020; Boolaky and Quick 2016; Seebeck 2024; Jahan and Karim 2024; Pelzer 2021). This ongoing debate underscores the need for further research to assess the practical benefits of CAMs for stakeholders, including regulators. In particular, whether and the extent to which CAM reporting aligns with the SEC's filing review remains less explored.²

Governed by Section 408 of the Sarbanes–Oxley Act (SOX) (SEC 2019), the SEC's review of regulatory filings aims to enhance compliance with disclosure rules and regulations (Cassell et al. 2013; Heese et al. 2017). When deficiencies are identified, the SEC issues comment letters requesting clarification or additional information, a process commonly referred to as the resolution process, which reflects investors' priorities. Prior studies document that the SEC's filing review improves financial reporting quality (e.g., Cunningham and Leidner 2022). Fang et al. (2024) argue that the SEC, as an independent authority responsible for interpreting applicable accounting standards, plays a stronger monitoring role than auditors.

Although external audits and the SEC's filing review represent two independent mechanisms of financial reporting monitoring, we hypothesize that CAMs are likely to identify issues that also appear in SEC comment letters. First, the purpose of CAM reporting is to inform investors by highlighting areas of significant auditor judgement and accounting policy uncertainty, areas that may also attract regulatory attention during the SEC's filing review. Second, prior research (e.g., Li et al. 2025) suggests that CAMs may be indicative of poor financial reporting quality, which could trigger heightened scrutiny from the SEC. Alternatively, auditors may report CAMs pre-emptively in anticipation of greater regulatory focus on complex or subjective matters, given the SEC's role as a robust financial reporting oversight body.³

We collect all CAM reports from Item 8 of 10-K filings, specifically within the 'Report of Independent Registered Public Accounting Firm' section. Within this section, we extract the text labelled 'Critical Audit Matter(s)'. Data on SEC comment letters related to 10-K filings are obtained from Audit Analytics. Our analysis focuses on the periods following the CAM mandate in 2019. The final sample consists of 997 firm-year observations from 870 unique firms with at least one SEC comment letter between 2019 and 2022.

To examine whether SEC comment letters address concerns aligned with those identified in auditors' CAM reports, we first cross-reference the 'Notes to financial statements' (hereafter

'Note') cited in both SEC comment letters and CAM reports. Specifically, we assess whether the Note numbers referenced in CAM reports are more likely to be mentioned in SEC comment letters, providing a direct link between the issues highlighted by auditors and those scrutinized by regulators. We find that SEC comment letters are more likely to reference the same Note numbers cited in CAM reports, suggesting alignment between the two. Furthermore, we find that firms with CAMs are more likely to involve an SEC supervisor during the resolution process. Prior studies suggest that the involvement of an SEC supervisor signals more substantive filing review and heightened regulatory attention (e.g., Heese et al. 2017; Ege et al. 2020). Our findings demonstrate that auditors' concerns closely align with those raised by the SEC.

Next, we conduct an accounting topic pair analysis between CAMs and SEC comment letters. Prior studies find that revenue recognition is the most frequently addressed issue in CAMs, followed by intangible assets and mergers and acquisitions (M&A).⁴ We document that firms with revenue-related CAMs are 59.4% more likely to receive SEC comment letters concerning revenue recognition than other issues. Similarly, firms with CAMs related to intangible assets and M&A are 130% and 78.6% more likely, respectively, to receive related comment letters.

Additionally, we analyze whether the uncertainty sentiment in CAM reports is associated with the likelihood of SEC comment letters addressing the same issues. We posit that higher uncertainty in CAMs increases the likelihood that SEC comment letters will raise the same concerns highlighted by auditors. Prior studies document that the tone or sentiment of disclosures reflects issue severity and influences how financial statement users respond (e.g., Allee and DeAngelis 2015; Frankel et al. 2021; Henry and Leone 2016). Higher uncertainty sentiment in CAMs may reflect unresolved risks or significant uncertainties, which are likely to align with the SEC's filing review priorities. For our analysis, we measure uncertainty sentiment using a masked language model. Specifically, we employ the RoBERTa (Robustly Optimized BERT Pre-training Approach) architecture, developed by researchers at Meta (formerly Facebook), using a self-supervised approach without human-labelled text.⁵ We utilize a fine-tuned version, optimized for uncertainty-related tasks. This model generates confidence scores (or probabilities) indicating the likelihood that each CAM belongs to the 'uncertainty' class. We separately measure the uncertainty sentiment of CAMs pertaining to revenue recognition, intangible assets and M&A and repeat the analysis. Consistent with our expectations, we find that firms with higher uncertainty sentiment in their CAMs are more likely to receive SEC comment letters addressing the same issues. Overall, our findings suggest a strong alignment between auditors' CAMs and issues raised in SEC comment letters.

This study makes two key contributions. First, it contributes to the ongoing debate regarding the usefulness of CAM reports for financial statement users. Since the adoption of AS 3101, the PCAOB has sought to evaluate the impact of CAM requirements on preparers, investors and audit firms (PCAOB 2020a). Notably, some audit engagement partners have expressed scepticism, stating that 'CAMs provide little value to investors' (PCAOB 2020b). Prior research offers mixed evidence on the usefulness of CAM reports. For example, Klevak et al. (2023)

and Burke et al. (2023) find limited evidence of negative market reactions to CAM reports; other studies generally find no significant investor response (e.g., Lennox et al. 2023; Liao et al. 2023). We document that auditors and regulators focus on similar areas of financial reporting complexity and uncertainty. This alignment between CAMs and SEC comment letters (1) supports the informational validity of CAMs by highlighting substantive financial reporting issues relevant to investors and (2) enhances the informational credibility of CAMs in the eyes of market participants.

Second, we contribute to the literature on SEC comment letters. The alignment between CAMs and comment letters implies that CAMs may serve as a proxy for regulatory attention. Our evidence suggests that these two monitoring mechanisms can reinforce each other by focusing on similar areas of financial reporting uncertainty and complexity. This alignment indicates that CAMs provide meaningful signals of substantive issues, underscoring their relevance within the context of regulatory oversight.

2 | Background and Hypothesis Development

2.1 | SEC Comment Letters and CAM Reporting Requirements

The nature and scope of SEC filing review are determined by the criteria outlined in Section 408 of SOX, along with other factors identified in its selective review criteria (SEC 2019). The review process aims to enhance compliance with applicable disclosure rules and regulations while improving reporting quality (Cassell et al. 2013; Heese et al. 2017; Cunningham and Leidner 2022). SEC comment letters are a key component of the SEC's regulatory filing review (Johnston and Petacchi 2017), seeking clarification, additional information and, in some cases, amendments or restatements to filings. Johnston and Petacchi (2017) further document that the comment letter resolution is associated with reduced information asymmetry and higher earnings response coefficients.

The CAM requirements are specified in Paragraphs 11–17 of PCAOB AS 3101: *The Auditor's Report on an Audit of Financial Statements When the Auditor Expresses an Unqualified Opinion* (PCAOB 2017). This standard took effect for audits of fiscal years ending on or after 30 June 2019, for large accelerated filers, and 15 December 2020, for all other companies. The CAM requirements aim to provide critical audit-specific information to bridge the information gap between investors and public companies, enabling better risk evaluation and informed investment decisions (PCAOB 2024). External auditors are required to provide a draft of the auditor's report and discuss it with the audit committee, addressing any disagreements with management, significant audit challenges and issues vital to financial reporting. The auditor's report must identify the CAMs (or state that none were identified) and explain the key considerations and how the CAMs were addressed. Appendix 1a provides an example of a CAM report for Universal Display Corp. in its 10-K filing for the fiscal year 2019. In this CAM report, KPMG references Notes 2 and 19 of the consolidated financial statements and identifies the

assessment of the estimated per-unit fee for long-term OLED contracts as a CAM due to the significant auditor judgement required to evaluate the estimates of total material units to be sold and royalties to be earned.

According to the PCAOB's investor survey, more than half of the investors believe that CAMs enhance their understanding of company and auditor disclosures. Specifically, investors noted that they would use CAMs to identify risks related to a company (66%), concentrate on key reporting issues (43%), analyse and compare companies for investment purposes (22%) and evaluate the quality of a company's audit (22%) (PCAOB 2020b). Croteau, the former deputy chief accountant at the SEC, remarked, 'In the context of a CAM, I believe there is informational value in highlighting those aspects that are most challenging, subjective, and complex' (Thomson Reuters 2023).

Extant literature examines the impact of CAM reports on audit and financial reporting quality. On the one hand, some studies find that CAM reports enhance both audit and financial reporting quality by increasing the perceived likelihood of PCAOB inspections or the auditor's internal reviews (e.g., Whitfield et al. 2024; Drake et al. 2024). Furthermore, Burke et al. (2023) document a significant increase in financial statement footnotes when these footnotes are referenced in CAM reports. Li et al. (2025) conduct a detailed analysis of CAMs, finding that revenue-related CAMs and tax-related CAMs are associated with improvements in revenue and tax accrual quality, respectively. On the other hand, some studies suggest that CAM reports negatively impact financial reporting and audit quality. For example, firms informed of impending CAMs by a close auditor tend to adopt more aggressive accounting estimates (Tan and Yeo 2022). Kachelmeier et al. (2020) find that CAM reports can reduce perceptions of auditor responsibility for future misstatements. Drake et al. (2024) find a modest but short-lived effect of tax-related CAM reports on tax-related financial reporting.

2.2 | Hypothesis Development

The auditor's CAM reporting requirements and the SEC's filing review represent two independent monitoring mechanisms within the financial reporting ecosystem. Although the SEC holds the ultimate authority in interpreting accounting standards and its regulatory filing review serves as a crucial safeguard for financial reporting quality, auditors primarily ensure compliance with applicable accounting standards (Fang et al. 2024). Auditors report CAMs in the client firm's 10-K filing, which is then subject to SEC filing review under the criteria set forth in SOX. Thus, alignment between issues identified in CAM reports and those raised in SEC comment letters suggests that both mechanisms reinforce each other by focusing on similar areas of financial reporting uncertainty (Bushman and Smith 2001; DeFond and Zhang 2014). This alignment also underscores the potential informational relevance of CAMs to investors.

CAMs aim to inform investors about risks related to financial reporting, and the SEC is likely to prioritize these areas during its review, given their increased importance to investor

protection. CAMs provide insights uniquely available from the auditor's perspective that are otherwise inaccessible to investors and other stakeholders (SEC 2017). If CAMs effectively highlight poor financial reporting quality or signal factors that may increase reporting risk, they may prompt closer scrutiny by the SEC during its filing review. Accordingly, we hypothesize that firms disclosing CAMs are more likely to receive SEC comment letters addressing the same or similar issues.

It is also possible that CAMs facilitate regulatory monitoring. In interviews with senior audit partners, Whitfield et al. (2024) argue that because CAMs represent items with the greatest risk of misstatement, audit firms and regulators such as the PCAOB are likely to focus more attention on these identified issues during their inspections. Chan and Liu (2023) use an analytical model demonstrating that CAM reports increase investor scrutiny. Their theoretical framework suggests that the effect of CAM reports on investor scrutiny depends on auditor liability and the clarity of the CAM signal. They further propose that auditor effort and investor scrutiny act as strategic substitutes, implying that heightened investor scrutiny in response to CAM reports occurs only when the CAM signal is sufficiently clear.

Budgetary and resource constraints at the SEC may also contribute to the alignment between CAM reports and SEC comment letters. Although the SEC ideally aims to conduct thorough filing reviews uniformly across all firms, resource constraints often render this goal unfeasible (e.g., Gunny and Hermis 2019; Ege et al. 2020; Kedia and Rajgopal 2011). As a result, the SEC may strategically concentrate its regulatory efforts on accounts and disclosures that present elevated financial reporting risks already identified by auditors in CAM reports. Accordingly, CAMs can serve as a valuable tool for the SEC, allowing it to focus its regulatory efforts more efficiently on areas of concern.

On the other hand, auditors may identify CAMs in anticipation of heightened regulatory scrutiny on issues involving highly subjective or complex judgements. Prior research highlights the critical role of regulatory oversight in shaping auditors' incentives and competencies (e.g., DeFond and Zhang 2014). Fang et al. (2024) argue that auditors are influenced by SEC scrutiny, given the SEC's more robust disciplinary role relative to auditors themselves. The anticipation of increased SEC monitoring on complex matters may therefore strengthen auditors' incentives to report CAMs.

In Appendix 1b, we present an example of an SEC comment letter issued to Universal Display Corp. regarding its 10-K filing for the fiscal year 2019. In Item 2 of the comment letter, the SEC refers to Note 19, which was also mentioned in the CAM report in the same 10-K filing (as shown in Appendix 1a), and requests clarification on the company's approach to disclosing disaggregated revenue information. This inquiry arises from the fact that the firm recognizes revenue based on the material units sold at the estimated per-unit fee, which is the same concern highlighted in the auditor's CAM report. In Appendix 2, we present another SEC comment letter that asks the firm to disclose how CAMs are addressed during the audit. This comment letter suggests that the SEC considers the auditor's remediation procedures as part

of its regulatory filing review process. Therefore, we state our first hypothesis:

Hypothesis 1a. *Firms that report CAMs are more likely to receive SEC comment letters concerning the same corresponding issue.*

Extant literature documents that the uncertainty sentiment of corporate disclosures affects market reactions. For example, Loughran and McDonald (2011) find that the market reacts more negatively to 10-K filings with a higher level of uncertainty sentiment. Given the SEC's mandate to protect investors, it is likely that increased uncertainty sentiment in CAMs signals heightened concerns from auditors, thereby aligning with the SEC's regulatory focus. As a result, the SEC may be more inclined to issue comment letters including those issues with higher uncertainty sentiment in CAM reports. Therefore, we state our second hypothesis:

Hypothesis 1b. *The uncertainty sentiment in the auditor's CAM report is positively associated with the likelihood of SEC comment letter issuance concerning the same corresponding issue.*

3 | Research Design

For Hypothesis 1a, we examine the alignment between concerns raised in CAMs and SEC comment letters using two tests: (1) assessing whether SEC comment letters are more likely to reference the same Note numbers cited in CAMs and (2) examining SEC filing reviews that involve a supervisor more frequently when CAMs are disclosed. In addition, we conduct an accounting topic pair analysis, focusing on three key issues commonly addressed in CAM reports: revenue recognition, intangible assets and M&A. We examine whether SEC comment letters are more likely to reference these topics when they are mentioned in CAMs. Our sample consists of 1087 firm-year observations with at least one SEC comment letter during the sample period.⁶

For Hypothesis 1b, we investigate whether the likelihood of raising a similar issue in comment letters is higher as the uncertainty sentiment in CAM reports regarding that issue rises. To measure uncertainty sentiment, we use a masked language model to analyse the three accounting topics in CAM reports. We provide a detailed description of the model in Section 3.2.1.

3.1 | Sample Selection

We collect data for our analyses from the SEC EDGAR (for CAMs), Audit Analytics (for comment letters) and Compustat/CRSP (for financial and market data). We first begin by collecting all comment letters. Since AS 3101 took effect for large accelerated filers for fiscal years ending on or after 30 June 2019 and for all other companies for fiscal years ending on or after 15 December 2020, we collect our sample beginning with fiscal year 2019. We obtain data on SEC comment letters issued for 10-K filings from 2019 to 2022 from Audit Analytics, as CAMs are disclosed in the auditor's report within 10-K filings, which are also the primary focus of the SEC's regulatory filing review (Bozanic et al. 2017).

TABLE 1 | Sample selection and distribution of firms.

Panel A. Sample selection			
Sample selection			SEC comment letter analysis
Firm-years in Compustat excluding non-US firms between 2019 and 2022			28,212
Less firm identifier unmatched with SEC comment letter data			(6430)
Less firm-years with missing data for control variables			(12,911)
Less firm-years with no observation for SEC comment letters			(7784)
Less emerging growth companies not subject to CAM requirements			(90)
Final sample for SEC comment letters analysis			997
Less firm-years without CAM			(42)
Final sample for accounting topic pair analysis			955
Panel B. Distribution of firms by year			
Year	Firms receiving SEC comment letters	Firms with CAMs	Proportion of firms with CAMs
2019	199	193	96.98%
2020	188	183	97.34%
2021	344	330	95.93%
2022	266	249	93.61%
Total	997	955	95.79%
Panel C. Distribution of firms by industry			
	Firms receiving SEC comment letters	Firms with CAMs	Proportion of firms with CAMs
Industrial Applications and Services	99	96	96.97%
Office of Energy & Transportation	165	157	95.15%
Office of Real Estate & Construction	33	31	93.94%
Office of Manufacturing	202	194	96.04%
Office of Technology	169	164	97.04%
Office of Trade & Services	229	220	96.07%

(Continues)

TABLE 1 | (Continued)

Panel C. Distribution of firms by industry			
	Firms receiving SEC comment letters	Firms with CAMs	Proportion of firms with CAMs
Office of Finance (Crypto Assets)	14	14	100.00%
Office of Life Sciences	86	79	91.86%
Total	997	955	95.79%

Note: In Table 1, Panel A, we summarize the sample selection procedure. Panel B presents the resultant distribution of these firms subject to the CAM mandate by year. In Panel C, we present the distribution of firms by the SEC Industry Office.

We collect all CAM reports in the 10-K filings available on the SEC EDGAR. Specifically, we start by gathering the ‘Report of Independent Registered Public Accounting Firm’ section in Item 8 of the 10-K filings. Then, we extract the section labelled ‘Critical Audit Matter(s)’. When auditors conclude that no CAMs exist, they are required to include the statement ‘We determined that there are no critical audit matters’ in their report. Consistent with AS 3101.05, we exclude Emerging Growth Companies, as they are not subject to CAM reporting requirements (Li et al. 2025).⁷

In our analysis of accounting topics (for Hypothesis 1a), we focus on the three primary areas identified most often by auditors in previous research: revenue recognition, intangible assets and M&A (Burke et al. 2023; Li et al. 2025). We specifically look for keywords in the CAM reports: ‘revenue’ and ‘sales return’ for revenue-related CAMs; ‘intangible’, ‘franchise’, ‘copyright’, ‘licence’, ‘patent’, ‘trademark’ and ‘royalty’ for CAMs concerning intangible assets; and ‘mergers and acquisition(s)’ or ‘business combination(s)’ for those related to M&A.^{8,9}

Our final sample comprises 997 firm-year observations from 870 unique firms with at least one SEC comment letter. Of these, 955 observations (95.79%) correspond to firms with at least one CAM, whereas 42 observations (4.21%) are for firms without CAMs. The sample selection procedure is summarized in Table 1, Panel A.¹⁰ In Panel B, we present the sample distribution by year. We note that the proportion of firms reporting CAMs is relatively stable over time, ranging from 93.61% to 97.34%.¹¹ In Panel C, we present the sample distribution by SEC industry office. Firms reviewed by the Office of Finance (Crypto Assets) exhibit the highest proportion of CAM reports at 100%, whereas the Office of Life Sciences shows the lowest proportion at 91.86%.

3.2 | Variable Construction

3.2.1 | CAMs and Related Variables

CAM is an indicator variable set to 1 if a firm discloses an auditor’s report with any descriptions of CAMs in year t and 0 if a firm does not report any CAMs. Among firms with CAMs, we create indicator variables for the analysis of Hypothesis 1a: (1)

CAM_Revenue, set to 1 if a firm reports CAMs pertaining to revenue recognition; (2) *CAM_Intangible*, set to 1 if a firm reports CAMs concerning intangible assets; and (3) *CAM_M&A*, set to 1 if a firm reports CAMs related to M&A.

CAM treats all CAM reports as homogeneous, equating the existence of CAM reports with the uncertainties expressed by the auditor. To capture the extent of a firm's accounting policy uncertainties, we measure uncertainty sentiment using a masked language model for the analysis of Hypothesis 1b.¹² Specifically, we employ the RoBERTa (Robustly Optimized BERT Pre-training Approach) architecture provided by the Hugging Face team. Built on Google's deep learning BERT (Bidirectional Encoder Representations from Transformers) language model, RoBERTa was developed by researchers at Meta using a self-supervised approach without human-labelled text.¹³ We utilize a fine-tuned version of Hugging Face's model hub, optimized for uncertainty-related tasks. This model generates confidence scores (or probabilities) indicating the likelihood that each CAM belongs to the 'uncertainty' class.

In Appendix 3a, we present an example of a CAM report from Frequency Electronics Inc. for the fiscal year ending 30 April 2021, which has a low uncertainty sentiment score based on the RoBERTa model. The report simply states the method used for recognizing revenue. In Appendix 3b, we present a CAM report from IMAX Corporation for the fiscal year ending 31 December 2021, with a high uncertainty sentiment score. This report discusses the uncertainties related to determining a sale or lease, the transaction price, the allocation of the transaction price and the timing of revenue recognition. We note that the RoBERTa model effectively captures the *context* in which uncertainties originate rather than relying on uncertainty-related keywords.

For our analysis, we measure the uncertainty sentiment of CAMs related to revenue recognition, intangible assets and M&A, separately (*Uncertainty_Revenue*, *Uncertainty_Intangible* and *Uncertainty_M&A*, respectively).

3.2.2 | SEC Comment Letter Variables

CL_Referenced is an indicator variable set to 1 if the comment letter seeks clarification on specific Note numbers in the financial statements that are referenced in the auditor's CAM report and 0 if the comment letter does not reference specific Note numbers in the CAM report.¹⁴ In other words, for firms that have received SEC comment letters, we examine whether the Note numbers addressed in the letter correspond to those referenced in the CAM report. If SEC staff increase regulatory scrutiny over specific Note numbers referenced in the CAM report during the filing review, we expect the resulting comment letters to more likely include discussions related to those corresponding Note numbers (as detailed in Appendix 1b).

Within firms that have received an SEC comment letter, we also capture the level of SEC involvement by examining whether the comment letter signatory is an SEC supervisor (*CL_Supervisor*).¹⁵

Prior studies argue that the involvement of SEC supervisors reflects more substantive filing review and greater regulatory attention (e.g., Heese et al. 2017; Ege et al. 2020).

In addition, we examine specific topics raised in the comment letters. Following the CAM topics, we assess whether the comment letter addresses concerns related to revenue recognition (*CL_Revenue*), intangible assets (*CL_Intangible*) and M&A (*CL_M&A*), as defined by Audit Analytics. We note that there can be multiple rounds of exchanges between the SEC and a firm until all issues are resolved. For our analysis, we focus on the first comment letters issued by the SEC, as these letters generally encompass the full scope of disclosure deficiencies identified by the agency (Ege et al. 2020).¹⁶

3.3 | Model

Our sample procedure yields a sample of firms with at least one SEC comment letter. Then, we investigate whether the SEC comment letter addresses specific Note numbers or accounting topics that are referenced in CAMs of the auditor's report. By presenting the direct link between CAM reports and SEC comment letters, we alleviate the endogeneity concern that the issuance of SEC comment letters stems from factors unrelated to the auditor's concern. We also note that CAM firms and non-CAM firms differ systematically across observable firm characteristics.¹⁷ To address this issue, we employ entropy balancing to balance the control variables between CAM and non-CAM firms for all analyses in this study.¹⁸ Specifically, we construct the entropy-balanced sample using the vector of control variables in Equation (1) while controlling for residual macro and micro factors through industry and year matching.

We first examine the association between CAM reports and SEC comment letters using the entropy-balanced sample. We use the following regression model:

$$\begin{aligned} DV_{i,t} = & a_0 + a_1 CAM_{i,t} + a_2 Size_{i,t} + a_3 IC_Weak_{i,t} + a_4 Restatement_{i,t} \\ & + a_5 HighVolatility_{i,t} + a_6 Age_{i,t} + a_7 Loss_{i,t} + a_8 ZScore_{i,t} + a_9 Chg_Sales_{i,t} \\ & + a_{10} M\&A_{i,t} + a_{11} Restructure_{i,t} + a_{12} External_Financing_{i,t+1} + a_{13} Big4_{i,t} \\ & + a_{14} 2Tier_{i,t} + a_{15} Segment_{i,t} + Year_FE + SEC\ Industry\ Office_FE + \epsilon. \end{aligned} \quad (1)$$

Our first two dependent variables (*DV*) are *CL_Referenced* and *CL_Supervisor*. *CL_Referenced* is set to 1 if an SEC comment letter issued for year *t*'s 10-K filing addresses the same Note numbers referenced in the auditor's CAM report of the client firm's 10-K filing for year *t*. *CL_Supervisor* is set to 1 if an SEC comment letter issued for year *t*'s 10-K filing is written by an SEC supervisor. To the extent that the SEC increases regulatory scrutiny over the same issues that were raised in the auditor's CAM report, we expect a higher likelihood of addressing issues referenced in the auditor's CAM report and a higher likelihood of involving SEC supervisors (i.e., $a_1 > 0$).

Our next three dependent variables are an indicator variable set to 1 if the SEC issues comment letters regarding revenue

recognition (*CL_Revenue*), intangible assets (*CL_Intangible*) and M&A (*CL_M&A*) for year *t*'s 10-K filing. Consistently, we expect to find significantly positive associations between the topics of the comment letters and the CAM reports on these three accounting areas.

Our treatment variable is an indicator variable set to 1 if a firm reports CAMs in the auditor's report (*CAM*) in its 10-K filing for year *t*. For the accounting topic pair analysis, we replace *CAM* with an indicator variable set to 1 if a firm discloses CAMs pertaining to revenue recognition (*CAM_Revenue*), intangible assets (*CAM_Intangible*) and M&A (*CAM_M&A*). Definitions of all variables are provided in Appendix 4.¹⁹

Additionally, we anticipate that the likelihood of receiving SEC comment letters will increase as the level of uncertainty sentiment expressed in the auditor's report rises. To test this, we

substitute *CAM_Revenue*, *CAM_Intangible* and *CAM_M&A* with *Uncertainty_Revenue*, *Uncertainty_Intangible* and *Uncertainty_M&A*, respectively, in Equation (1).

To account for macroeconomic and industry-specific factors that are not explicitly modelled, we include SEC Industry Office and year fixed effects.²⁰ For our binary dependent variables, we estimate the models using a linear probability model to mitigate incidental parameter concerns in panel data with multiple fixed effects (Greene 2004).

Following previous studies (e.g., Cassell et al. 2013; Kolev et al. 2023), we select the control variables in Equation (1) to analyse the factors influencing the issuance of SEC comment letters, based on the criteria outlined in Section 408 of SOX. Specifically, we consider the quality of internal controls and financial restatements (*IC_Weak* and *Restatement*), firm size

TABLE 2 | Descriptive statistics.

Variable	N	Mean	SD	p1	p25	p50	p75	p99
CL_Referenced	997	0.042	0.201	0.000	0.000	0.000	0.000	1.000
CL_Supervisor	997	0.016	0.126	0.000	0.000	0.000	0.000	1.000
CL_Revenue	955	0.142	0.350	0.000	0.000	0.000	0.000	1.000
CL_Intangible	955	0.082	0.274	0.000	0.000	0.000	0.000	1.000
CL_M&A	955	0.111	0.314	0.000	0.000	0.000	0.000	1.000
CAM	997	0.958	0.201	0.000	1.000	1.000	1.000	1.000
CAM_Revenue	955	0.301	0.459	0.000	0.000	0.000	1.000	1.000
CAM_Intangible	955	0.072	0.259	0.000	0.000	0.000	0.000	1.000
CAM_M&A	955	0.172	0.377	0.000	0.000	0.000	0.000	1.000
Uncertainty_Revenue	955	0.156	0.238	0.000	0.000	0.000	0.517	0.521
Uncertainty_Intangible	955	0.037	0.134	0.000	0.000	0.000	0.000	0.519
Uncertainty_M&A	955	0.089	0.196	0.000	0.000	0.000	0.000	0.521
Size	955	8.037	2.067	2.306	6.864	8.098	9.366	11.900
Restatement	955	0.013	0.111	0.000	0.000	0.000	0.000	1.000
IC_Weak	955	0.103	0.304	0.000	0.000	0.000	0.000	1.000
HighVolatility	955	0.186	0.390	0.000	0.000	0.000	0.000	1.000
Age	955	2.967	0.917	0.693	2.197	3.178	3.638	4.277
Chg_Sales	955	0.226	0.705	-0.670	-0.001	0.105	0.265	3.394
Loss	955	0.328	0.470	0.000	0.000	0.000	1.000	1.000
M&A	955	0.438	0.496	0.000	0.000	0.000	1.000	1.000
Restructure	955	0.014	0.116	0.000	0.000	0.000	0.000	1.000
ZScore	955	5.020	2.253	1.000	3.000	5.000	7.000	10.000
Segment	955	2.211	0.820	0.000	1.792	2.303	2.773	3.829
External_Financing	955	0.005	0.192	-0.388	-0.069	-0.018	0.023	0.730
Big4	955	0.823	0.382	0.000	1.000	1.000	1.000	1.000
2Tier	955	0.099	0.299	0.000	0.000	0.000	0.000	1.000

TABLE 3 | The association between CAM reports and SEC comment letters.

Panel A. Covariate balance for entropy balanced sample partitioned on CAM						
Variable	Mean		Standardized differences	Variance		Variance ratios
	CAM = 0	CAM = 1		CAM = 0	CAM = 1	
Size	7.215	7.215	0.000	4.615	6.065	1.314
Restatement	0.016	0.016	0.000	0.015	0.015	0.999
IC_Weak	0.131	0.131	0.000	0.114	0.114	0.999
HighVolatility	0.209	0.209	0.000	0.165	0.165	0.999
Age	2.869	2.869	0.000	1.134	0.877	0.773
Chg_Sales	0.210	0.210	0.000	0.446	0.617	1.382
Loss	0.421	0.421	0.000	0.244	0.244	1.000
M&A	0.382	0.382	0.000	0.236	0.236	1.000
Restructure	0.015	0.015	0.000	0.015	0.015	0.999
ZScore	5.207	5.207	0.000	8.488	6.599	0.777
Segment	2.081	2.081	0.000	0.833	0.769	0.923
External_Financing	0.055	0.055	0.000	0.095	0.082	0.859
Big4	0.719	0.719	0.000	0.202	0.202	1.000
2Tier	0.123	0.123	0.000	0.108	0.108	0.999
Panel B. Regression						
Variables	CL_Referenced		CL_Supervisor			
	(1)	(2)				
CAM	0.080***	0.020**				
	(5.048)	(2.124)				
Size	−0.003	−0.001				
	(−0.653)	(−0.690)				
Restatement	0.001	−0.011*				
	(0.092)	(−1.831)				
IC_Weak	−0.039***	−0.019***				
	(−2.588)	(−2.823)				
HighVolatility	0.013	−0.012				
	(0.817)	(−1.415)				
Age	0.014	−0.005				
	(1.533)	(−1.281)				
Chg_Sales	−0.004	0.003				
	(−0.686)	(0.850)				
Loss	0.010	0.014				
	(0.530)	(1.398)				
M&A	0.015	0.002				
	(1.255)	(0.247)				

(Continues)

TABLE 3 | (Continued)

Panel B. Regression		
	CL_Referenced	CL_Supervisor
Variables	(1)	(2)
Restructure	−0.035*** (−2.944)	0.052 (0.673)
ZScore	−0.001 (−0.384)	0.000 (0.007)
Segment	−0.001 (−0.137)	0.005 (0.808)
External_Financing	0.025 (1.096)	0.006 (0.536)
Big4	−0.024 (−0.793)	0.009 (1.118)
2Tier	0.011 (0.345)	0.013 (1.290)
Year FE	Yes	Yes
Industry FE	Yes	Yes
SE clustered by firm	Yes	Yes
Observations	997	997
Adj R^2	0.019	0.031

Note: In Panel A, we present the entropy-balanced sample comparisons between firms with CAMs and those without. The standardized differences are calculated as the difference in means between the treatment and control samples, scaled by the standard deviation of the respective variable in the treatment sample for each covariate. The variance ratios are determined by dividing the standard deviation of each covariate in the treatment sample by that in the control sample. In Panel B, in Column 1, the dependent variable is an indicator set to 1 if the SEC comment letter addresses specific Note numbers referenced in the auditor's CAM report and 0 if it does not. In Column 2, the dependent variable is an indicator set to 1 if an SEC supervisor is involved during the comment letter resolution process and 0 otherwise. We estimate the model using a linear probability model (LPM) with standard errors clustered by firm. *, ** and *** indicate two-tailed significance at the 10%, 5% and 1% levels, respectively; *t*-statistics are presented in parentheses below the coefficients. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Appendix 4.

(Size) and stock price volatility (*HighVolatility*). Recognizing the SEC's discretion in selecting firms for regulatory filing review, as detailed in SOX, we expand the control variables based on prior research and include firm age (*Age*), an indicator variable for net income loss (*Loss*), financial distress (*ZScore*), sales growth (*Chg_Sales*), the total amount of equity and debt financing in year $t + 1$ (*External_Financing*), an indicator variable for M&A (*M&A*) and an indicator variable for restructuring activities (*Restructure*), as well as indicators for Big 4 auditors (*Big4*) and second-tier auditors (*2Tier*), along with the number of business and geographic segments (*Segment*).

4 | Results

4.1 | Descriptive Statistics

In Table 2, we present the descriptive statistics. We observe that 95.8% of firms disclose at least one CAM. Within firms that report at least one CAM, we find that 30.1% pertain to revenue recognition, 7.2% to intangible assets and 17.2% to M&A during the sample period. Regarding the uncertainty sentiment of CAMs, we note that, on average, revenue-CAMs, intangible assets-CAMs

and M&A-CAMs exhibit uncertainty scores of 0.156, 0.037 and 0.089, respectively, based on the RoBERTa model.

Turning to SEC comment letters, we find that 4.2% of SEC comment letters reference specific Note numbers mentioned in the CAM reports.²¹ We also observe that an SEC supervisor is involved in 1.6% of SEC comment letter cases. Within firms having received an SEC comment letter, 14.2% address concerns related to revenue recognition, 8.2% pertain to intangible assets, and 11.1% focus on M&A.

4.2 | The Relation of CAM Reports With SEC Comment Letters

4.2.1 | CAM Report Analysis

In Table 3, Panel A, we present the covariate balance between CAM and non-CAM firms. We observe that the standardized differences and variance ratios of the covariates are comparable between the two groups, indicating the effectiveness of the balancing procedure. Panel B presents the results from estimating Equation (1). Consistent with our expectations, we

TABLE 4 | Accounting topic pair analysis.

Variables	Dependent variable		
	CL_Revenue	CL_Intangible	CL_M&A
	(1)	(2)	(3)
CAM_Revenue	0.084*** (2.981)		
CAM_Intangible		0.107** (2.040)	
CAM_M&A			0.087** (2.531)
Size	−0.007 (−0.960)	−0.001 (−0.149)	−0.014** (−2.047)
Restatement	−0.184*** (−5.697)	−0.103*** (−3.276)	−0.033 (−0.378)
IC_Weak	−0.023 (−0.509)	−0.007 (−0.208)	0.052 (1.142)
HighVolatility	0.058* (1.794)	0.022 (0.910)	0.034 (1.175)
Age	−0.017 (−1.077)	0.001 (0.046)	0.005 (0.296)
Chg_Sales	0.019 (0.848)	−0.003 (−0.368)	0.015 (0.786)
Loss	0.009 (0.300)	−0.004 (−0.170)	−0.057** (−2.096)
M&A	0.028 (1.182)	0.043** (2.220)	0.012 (0.558)
Restructure	−0.037 (−0.463)	−0.020 (−0.262)	0.017 (0.169)
ZScore	0.004 (0.661)	0.007 (1.579)	0.001 (0.095)
Segment	0.028 (1.646)	0.016 (1.218)	0.023 (1.341)
External_Financing	−0.020 (−0.345)	0.054 (1.030)	0.138** (2.148)
Big4	−0.062 (−1.085)	−0.001 (−0.035)	0.044 (0.884)
2Tier	−0.028 (−0.443)	0.033 (0.675)	0.081 (1.353)
Year FE	Yes	Yes	Yes

(Continues)

TABLE 4 | (Continued)

Variables	Dependent variable		
	CL_Revenue	CL_Intangible	CL_M&A
	(1)	(2)	(3)
Industry FE	Yes	Yes	Yes
SE clustered by firm	Yes	Yes	Yes
Observations	955	955	955
Adj R^2	0.051	0.042	0.021

Note: The dependent variables in Columns 1, 2 and 3 are an indicator variable set to 1 if a firm receives an SEC comment letter addressing revenue recognition, intangible assets and M&A issues, respectively. The models are estimated using LPM, with standard errors clustered by firm. *, ** and *** indicate two-tailed significance at the 10%, 5% and 1% levels, respectively; t -statistics are presented in parentheses below the coefficients. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Appendix 4.

find a significantly positive association between *CAM* and *CL_Referenced* (t -statistic = 5.048), suggesting that SEC comment letters are more likely to address issues related to specific Note numbers referenced in the auditor's CAM report. The marginal effect of *CAM* in the fully specified model is 0.0804. In terms of economic significance, the estimated coefficient and sample statistics imply that when CAM reports discuss specific Note numbers, there is a 191% higher likelihood that the SEC will reference those corresponding Note numbers in the comment letters ($1 \times 0.0804 / 0.042 = 1.91$).

In Column 2, we find that the SEC comment resolution process is more likely to involve an SEC supervisor when a firm discloses an auditor's report with at least one CAM (t -statistic = 2.124) with the marginal effect of 0.0204, suggesting a higher level of SEC involvement. Economically, firms with CAMs experience a 128% greater likelihood of involving an SEC supervisor until all issues are resolved ($1 \times 0.0204 / 0.016 = 1.28$).

4.2.2 | Direct Link Analysis by Accounting Topics

In Table 4, we present a pairwise analysis of accounting topics between CAMs and SEC comment letters.²² In Column 1, we find that firms with revenue-related CAMs are more likely to receive SEC comment letters addressing issues related to revenue recognition, as indicated by a significantly positive estimated coefficient on *CAM_Revenue* (t -statistic = 2.981). Economically, the marginal effect suggests that firms with revenue-related CAMs have a 59.4% higher likelihood of receiving comment letters on revenue recognition from the SEC ($1 \times 0.0843 / 0.142 = 0.594$).

Similarly, we find significantly positive estimated coefficients for *CAM_Intangible* and *CAM_M&A* (t -statistics = 2.040 and 2.531, respectively). Economically, firms with CAMs related to intangible assets and M&A have a 130% and 78.6% higher likelihood of receiving SEC comment letters on these issues, respectively (intangible assets: $1 \times 0.1070 / 0.082 = 1.30$; M&A: $1 \times 0.0873 / 0.111 = 0.786$). Collectively, our evidence suggests an alignment between the issues raised in auditors' CAM reports and those addressed in SEC comment letters.²³

4.2.3 | Uncertainty Sentiment of CAM Report Analysis

In Table 5, we present the results for *Uncertainty_Revenue* (Column 1), *Uncertainty_Intangible* (Column 2) and *Uncertainty_M&A* (Column 3). Across all specifications, we find that the estimated coefficients are positive and statistically significant (t -statistics = 2.981, 2.043 and 2.535, respectively). These results suggest that the uncertainty sentiment expressed by the auditor increases the likelihood of the SEC issuing comment letters on the same corresponding accounting topic. Overall, our evidence across all specifications supports that auditors' uncertainty concerns regarding the client firm's accounting policies are positively associated with questions raised by SEC staff during regulatory filing reviews.

5 | Additional Analysis and Robustness Tests

5.1 | SEC EDGAR 10-K Filing Download as an Alternative Measure of Regulatory Monitoring

Our main analysis examines the association between the auditor's identification of CAMs and SEC monitoring through the issuance of comment letters. Although existing literature uses SEC comment letters to capture the outcome of SEC monitoring (e.g., Naughton et al. 2018; Kolev et al. 2023), concerns may arise regarding unobservable omitted variables, partly due to the private nature of the SEC's regulatory filing review. To address this concern, we utilize an alternative measure of external monitoring: EDGAR downloads of a firm's 10-K filings. The SEC provides an EDGAR log file dataset, which contains the EDGAR internet search traffic information that can be used to infer user access statistics. We build on the work of Stice-Lawrence (2023) and Iselin et al. (2024), who argue that SEC-initiated downloads of a firm's regulatory filings serve as a direct measure of the SEC's monitoring activities. Following this premise, we assess whether the number of downloads of 10-K filings on EDGAR is higher for firms with CAMs in the auditor's report. If financial statement users, including the SEC, increase their monitoring of firms with CAM reports, we expect the number of 10-K filing downloads for these firms to be greater than that of firms without CAM reports.

TABLE 5 | Uncertainty sentiment in CAM reports.

Variables	Dependent variable		
	CL_ Revenue	CL_ Intangible	CL_M&A
	(1)	(2)	(3)
Uncertainty_ Revenue	0.163*** (2.981)		
Uncertainty_ Intangible		0.207** (2.043)	
Uncertainty_M&A			0.169** (2.535)
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
SE clustered by firm	Yes	Yes	Yes
Observations	955	955	955
Adj R ²	0.051	0.042	0.021

Note: We use the Hugging Face natural language processing model to measure the uncertainty sentiment of CAM reports. We use the LPM model, with standard errors clustered by firm. *, ** and *** indicate two-tailed significance at the 10%, 5% and 1% levels, respectively; *t*-statistics are presented in parentheses below the coefficients. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Appendix 4.

For the analysis, we substitute the dependent variable in Equation (1) with the natural logarithm of the number of 10-K filing downloads on EDGAR for each firm (*SEC_Download*).²⁴ Consistent with the results from the comment letter analysis, we expect a positive estimated coefficient on *CAM*. In Table 6, we find a statistically significant positive association between *CAM* and *SEC_Download*, suggesting greater external monitoring for 10-K filings that include CAM reports.

5.2 | Falsification Tests

In the accounting topic pair analysis, we establish a direct link between the accounting topics referenced in CAM reports and the corresponding issues addressed in SEC comment letters. To strengthen our inferences, we conduct a falsification test. Specifically, we examine the association between three accounting topics identified in the CAM reports from our sample and other critical disclosure topics that differ from those covered in the CAM reports. Drawing on prior literature on SEC comment letters, we focus on major customer information disclosures (Ellis et al. 2012; Li et al. 2018) and segment reporting (Bens et al. 2011; Botosan and Stanford 2005). A lack of statistically significant associations would reduce concerns that our inferences are driven by spurious results.

Table 7 presents the findings. As expected, the estimated coefficients for *CAM_Revenue*, *CAM_Intangible* and *CAM_M&A*

TABLE 6 | SEC EDGAR 10-K filing downloads.

Variables	Dependent variable: SEC_Download
	(1)
CAM	0.129** (2.027)
Size	0.039*** (4.135)
Restatement	−0.372* (−1.823)
IC_Weak	−0.205*** (−3.639)
HighVolatility	0.078* (1.946)
Age	0.193*** (8.603)
Chg_Sales	−0.028 (−1.247)
Loss	−0.054 (−1.576)
M&A	−0.073*** (−2.697)
Restructure	0.017 (0.302)
ZScore	0.011 (1.495)
Segment	0.047** (2.237)
External_Financing	−0.051 (−0.649)
Big4	−0.233*** (−4.406)
2Tier	−0.163*** (−2.621)
Year FE	Yes
Industry FE	Yes
SE clustered by firm	Yes
Observations	791
Adj R ²	0.963

Note: The dependent variable is the natural logarithm of the number of 10-K filing downloads on SEC EDGAR. The models are estimated using OLS, with standard errors clustered by firm. *, ** and *** indicate two-tailed significance at the 10%, 5% and 1% levels, respectively; *t*-statistics are presented in parentheses below the coefficients. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Appendix 4.

TABLE 7 | Falsification test.

Variables	Dependent variable					
	CL_MajorCustomer			CL_Segment		
	(1)	(2)	(3)	(4)	(5)	(6)
CAM_Revenue	−0.004 (−0.441)			−0.023 (−0.810)		
CAM_Intangible		−0.005 (−0.334)			0.062 (1.135)	
CAM_M&A			0.018 (1.313)			0.029 (0.741)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
SE clustered by firm	Yes	Yes	Yes	Yes	Yes	Yes
Observations	955	955	955	955	955	955
Adj R^2	0.033	0.033	0.035	0.044	0.045	0.044

Note: The models are estimated using LPM, with standard errors clustered by firm. *, ** and *** indicate two-tailed significance at the 10%, 5% and 1% levels, respectively; t -statistics are presented in parentheses below the coefficients. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Appendix 4.

are both economically and statistically indistinguishable from 0 across all specifications.

We also acknowledge the possibility that auditors may include in their CAM reports topics that were previously raised in SEC comment letters, which could confound our findings if there is persistence in the topics over time. To address this concern, we conduct an additional falsification test by examining whether CAM reports in year t are associated with SEC comment letters on the same topics in year $t - 1$. This approach allows us to assess whether the documented association is driven by prior SEC scrutiny rather than contemporaneous alignment between auditors and regulators. We find that the estimated coefficients on *CAM_Revenue*, *CAM_Intangibles* and *CAM_M&A*, when measured with a 1-year lag, are not statistically significant. This untabulated result provides further support that our main findings are not driven by spurious correlations with prior SEC comment letter topics.

5.3 | Accounting Reporting Complexity

Despite including a comprehensive set of control variables in Equation (1), we recognize that a firm's inherent accounting reporting complexity (ARC) may confound the association between auditors' CAM reports and the SEC's comment letter issues concerning the same accounting topics. Firms with uniquely complex economic transactions and accounting practices may be more likely to receive SEC comment letters seeking clarification. The challenging and complex application of accounting standards in such cases may also prompt auditors to issue CAM reports highlighting areas requiring significant judgement, subjectivity or complexity.

To address this potential endogeneity concern, we include additional control variables in the model. Specifically, we include proxies for a firm's ARC as developed by Hoitash and Hoitash (2018). These proxies measure ARC by counting the number of accounting items tagged using eXtensible Business Reporting Language (XBRL) in a firm's 10-K filing. The authors argue that as financial reporting complexity increases, the number of XBRL tags for disclosed accounting concepts and items also increases.

Focusing on three specific accounting topics—revenue recognition, intangible assets and M&A—we compute ARC measures for each topic (*ARC_Revenue*, *ARC_Intangible* and *ARC_M&A*). These variables capture the reporting complexity pertaining to each respective area. Additionally, we calculate the ratio of XBRL tags pertaining to each topic to the total XBRL tags disclosed in a 10-K filing (*Ratio_Revenue*, *Ratio_Intangible* and *Ratio_M&A*). This ratio reflects the relative significance of each topic's reporting complexity within the overall financial report.

Table 8 presents the results. The estimated coefficients for *CAM_Revenue*, *CAM_Intangible* and *CAM_M&A* remain significantly positive, reinforcing our inferences. Moreover, the estimated coefficients for ARC and the ratio measures are insignificant, further supporting the robustness of our findings.

5.4 | Alternative Measure of CAM Uncertainty Sentiment

In Section 4.2.3, we employ the RoBERTa model to measure the uncertainty sentiment of CAM reports. To strengthen

TABLE 8 | Accounting reporting complexity.

Variables	Dependent variable		
	CL_Revenue	CL_Intangible	CL_M&A
	(1)	(2)	(3)
CAM_Revenue	0.086*** (3.043)		
ARC_Revenue	−0.015 (−0.601)		
Ratio_Revenue	−0.038 (−0.070)		
CAM_Intangible		0.098* (1.855)	
ARC_Intangible		0.001 (0.140)	
Ratio_Intangible		1.004 (1.113)	
CAM_M&A			0.083** (2.344)
ARC_M&A			0.007 (0.678)
Ratio_M&A			−0.075 (−0.161)
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
SE clustered by firm	Yes	Yes	Yes
Observations	955	955	955
Adj R^2	0.050	0.045	0.020

Note: The models are estimated using LPM, with standard errors clustered by firm. *, ** and *** indicate two-tailed significance at the 10%, 5% and 1% levels, respectively; t -statistics are presented in parentheses below the coefficients. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Appendix 4.

our findings, we repeat the analysis using a ‘bag-of-words’ approach. Specifically, we calculate the natural logarithm of the number of uncertainty-related keywords, as defined by Loughran and McDonald (2011), for revenue-related CAM reports, intangible asset-related CAM reports and M&A-related CAM reports separately (*Uncertainty_LM_Revenue*, *Uncertainty_LM_Intangible* and *Uncertainty_LM_M&A*, respectively).

In Table 9, we consistently find that the estimated coefficients remain significantly positive across all specifications. This suggests a higher likelihood of SEC comment letter issuance for accounting topics where the auditor expresses greater uncertainty concerns.

5.5 | Habitual CAM Reporter vs. Sporadic CAM Reporter

Beyer et al. (2025) provide descriptive evidence that some CAM topics (e.g., revenue from customer contracts) recur more frequently than other topics (e.g., business combinations) and that 68% of auditor reports include at least one recurring CAM topic. Consistently, we observe that a substantial portion of firms report CAMs consistently over the 4-year sample period: 22.58% never reported a CAM, and 15.67% reported CAMs once; 19.76%, twice; 20.17%, three times; and 21.82%, in all 4 years.

Following the recurring, intermittent and nonrecurring classification approach, we compare the likelihood of receiving SEC

TABLE 9 | Alternative uncertainty sentiment measure.

Variables	Dependent variable		
	CL_		
	CL_Revenue	Intangible	CL_M&A
	(1)	(2)	(3)
Uncertainty_ LM_Revenue	0.029** (2.494)		
Uncertainty_ LM_Intangible		0.034** (2.025)	
Uncertainty_ LM_M&A			0.030*** (2.603)
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Industry FE	Yes	Yes	Yes
SE clustered by firm	Yes	Yes	Yes
Observations	955	955	955
Adj R ²	0.049	0.041	0.022

Note: The models are estimated using LPM, with standard errors clustered by firm. *, ** and *** indicate two-tailed significance at the 10%, 5% and 1% levels, respectively; *t*-statistics are presented in parentheses below the coefficients. Continuous variables are winsorized at the 1% and 99% levels. All variables are defined in Appendix 4.

comment letters between firms that disclosed CAMs only once (*Sporadic_CAM*) and those that disclosed CAMs in all 4 years (*Habitual_CAM*). In univariate analysis, we find that *Habitual_CAM* firms exhibit a significantly higher likelihood of receiving SEC comment letters ($p < 0.01$), reinforcing our main findings (untabulated).

6 | Conclusion

This study examines the alignment between issues raised in auditors' CAM reports and SEC comment letters. Although these two monitoring mechanisms operate independently, our findings indicate that auditor-reported concerns correspond with those raised by the SEC. Specifically, (1) SEC comment letters are more likely to reference the same Note numbers cited in CAM reports, (2) there is a direct correspondence between the topics addressed in comment letters and those in CAM reports, and (3) firms exhibiting higher uncertainty sentiment in their CAMs are more likely to receive SEC comment letters on the same issues.

Our study underscores the relevance of CAM reports not only to investors but also to regulatory monitoring. Our findings provide further support for the informational value of CAM reports, an outcome of particular interest to investors. The alignment reinforces the informational validity of CAMs, demonstrating that they highlight substantive financial reporting issues relevant to investors. Moreover, since the adoption of the CAM mandate, the PCAOB has actively gathered evidence on the benefits and costs of CAM requirements through surveys of auditors and

investors, as well as interviews with audit committee chairs and preparers (PCAOB 2022). Our results lend support to the PCAOB's emphasis on CAMs' role in enhancing audit transparency and improving investor decision-making.

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Data Availability Statement

Data are available from the public sources cited in the text.

Endnotes

¹ On 23 October 2017, SEC Chairman Jay Clayton issued a statement supporting this initiative, noting that CAMs would 'provide investors and other financial statement users with the auditor's perspective on matters discussed with the audit committee that relate to material accounts or disclosures and involved especially challenging, subjective or complex auditor judgement' and therefore urged 'all involved in the implementation of the revised auditing standards, including the Commission and the PCAOB, to pay close attention to these issues going forward' (SEC 2017). For more on the SEC's perspective, Chairman Clayton's remarks on the critical role of audit committees in financial reporting can be found here: <https://www.sec.gov/newsroom/speeches-statements/statement-role-audit-committees-financial-reporting>.

² Using the UK data, Dwyer et al. (2024) find that auditor-identified risks align with those identified by audit committees. In contrast, US audit committees do not issue separate disclosures of such risks. Instead, management is required to disclose Critical Accounting Policies and Estimates in the Management's Discussion and Analysis section, where they address uncertainties involving significant judgement in applying accounting standards. Although the objectives of CAM and Critical Accounting Policies and Estimates disclosures differ, both relate to key judgements underlying financial reporting, suggesting potential overlap. This raises the question of whether CAMs provide incremental information relevant to investors. As our study focuses on the SEC's regulatory filing review, aimed at ensuring compliance with disclosure requirements, we leave this question to future research.

³ To clarify, we do not claim that the disclosure of CAMs in the auditor's report causes the issuance of SEC comment letters. Rather, we posit that the presence of CAMs may be associated with heightened scrutiny by SEC staff during the filing review process. Although CAMs and comment letters originate from independent mechanisms, their shared emphasis on accounting uncertainty and material disclosures may give rise to an observable association. CAMs may flag financial reporting areas that are likewise more likely to attract SEC attention. Supporting this notion, Ballesterio and Schmidt (2024) find that the involvement of more experienced auditors in the comment letter resolution process is associated with lower (higher) likelihood to restate (revise) prior filings, suggesting that the audit quality may share the outcome of the regulatory review.

⁴ Li et al. (2025) find that revenue-related CAMs account for 18.99% of all CAMs, followed by intangible assets-related CAMs at 15.87% and mergers and acquisitions-related CAMs at 11.75%. In contrast, Burke et al. (2023) report that 26.9% of issuers disclose CAMs related to intangible assets, followed by revenue-related CAMs at 26.1% and mergers and acquisitions-related CAMs at 22.4%.

⁵ We provide a detailed description of the model in section 3.2.1.

⁶ Our sample construction procedure may be overly restrictive, as we retain only firm-years with both SEC comment letters and CAM reports. We adopt this approach to mitigate endogeneity concerns by providing evidence of a direct link between (1) the specific Note

numbers referenced in CAMs and the corresponding Note numbers in SEC comment letters and (2) the accounting topics jointly raised in both disclosures. In untabulated analysis, we relax this sample restriction and examine the broader association between the presence of CAM reports and the likelihood of receiving an SEC comment letter on 10-K filings. Consistent with our main findings, we observe a positive and significant association, lending further support to our main inference that CAMs align with areas subject to regulatory scrutiny.

⁷ We manually inspect Page 1 of each Form 10-K filing to determine whether the firm self-identifies as an emerging growth company.

⁸ Burke et al. (2023) identify revenue-related CAMs based on ‘revenue from customer contracts’, ‘sales return and allowances’, ‘other revenue’ and ‘interest revenue’ topics; intangible-related CAMs based on ‘goodwill’, ‘other intangible assets’, ‘goodwill and intangible assets’ and ‘deferred and capitalized costs’ topics and M&A-related CAMs based on ‘business combination’ and ‘equity investments and joint ventures’ topics.

⁹ Our initial CAM sample consists of 14,978 firm-year observations, classify 3409 observations (22.76%) as related to revenue recognition, 905 observations (6.04%) as related to intangible assets and 2052 observations (13.70%) as related to M&A. The frequencies of CAMs related to revenue, intangible assets and M&A are consistent with those reported in prior studies (e.g., Burke et al. 2023; Klevak et al. 2023), supporting the effectiveness of the CAM topic classification method.

¹⁰ The frequency of SEC comment letter issuance is low and exhibits a declining trend in recent years. In our sample, 18% of firms received an SEC comment letter in 2019, followed by 8%, 13% and 12% in 2020, 2021 and 2022, respectively. This pattern is consistent with the findings of Cunningham and Leidner (2022), who document a steady decline in the frequency of firms receiving comment letters during the late 2010s, with the percentage dropping to 16% in 2017.

¹¹ In the first year of CAM adoption, we identify only six large accelerated filers that do not disclose CAMs in their auditor’s reports. Similarly, Burke et al. (2023) identify 10 issuers that report zero CAMs.

¹² We measure uncertainty sentiment solely within the auditor’s description of the CAMs. We exclude the section describing ‘procedures performed to address the CAMs’, as this section is typically boilerplate and standardized, often using generic language such as ‘evaluated’, ‘reviewed’, ‘inspected’ or ‘compared’.

¹³ For a detailed description of the model, see <https://huggingface.co/FacebookAI/roberta-base>. The model operates by pretraining a deep neural network on a large collection of text. Following pretraining, it is fine-tuned for specific natural language processing tasks, such as sentiment analysis. Once fine-tuned, the model can infer new text by leveraging its learned representations to make predictions. The RoBERTa model has exhibited superior performance in linguistic analysis than ‘bag of words’ approach in academic literature, including studies by Rawat and Singh Samant (2022) and Tan et al. (2022), published under the Institute of Electrical and Electronics Engineers (IEEE).

¹⁴ We use the acronym ‘CL’ to refer to comment letters issued by the SEC. We acknowledge that *CL_Referenced* would equal 0 if a firm does not report CAMs, according to the variable construct.

¹⁵ To identify SEC supervisors, we examine whether the signatories at the end of SEC comment letters hold titles such as assistant director, senior assistant or chief accountant. Prior studies use this measure to capture the intensity of SEC oversight and the level of SEC involvement in regulatory filing reviews (e.g., Cassell et al. 2013; Kolev et al. 2023). Accordingly, a positive association between CAMs and the likelihood of SEC comment letters issued by a supervisor suggests that firms reporting CAMs receive greater regulatory attention.

¹⁶ Other studies examining SEC comment letters also focus on the initial comment letters for their analyses (e.g., Kubic and Toynbee 2023; Cassell et al. 2019).

¹⁷ Before entropy balancing, we note that CAM and non-CAM firms differ statistically across the control variables in equation (1), with a $p < 0.05$, except for *Restatement* ($p = 0.477$).

¹⁸ McMullin and Schonberger (2020) state that entropy balancing outperforms other matching methods, like propensity score matching, by equalizing higher order moments of covariate distributions between the treatment and control groups. This method reduces both Type I and Type II errors, resulting in a more effective balancing of covariates. In our analysis, we balance the covariates on their mean, variance and skewness.

¹⁹ Section 408 of SOX requires the SEC to review each company at least once every 3 years. Because the identity of firms reviewed by SEC staff becomes publicly available only when an SEC comment letter is issued, the absence of a comment letter may be interpreted as follows: (1) The firm was not reviewed during the period, or (2) the firm was reviewed but the SEC did not identify disclosure deficiencies warranting a comment letter issuance. However, our study is not subject to this interpretive issue, as we focus on firms having received an SEC comment letter and examine whether the letter raises issues that are referenced in the auditor’s CAM reports.

²⁰ Detailed SIC codes assigned to each SEC office are available at <https://www.sec.gov/corpfin/division-of-corporation-finance-standard-industrial-classification-sic-code-list>.

²¹ We acknowledge a limitation of the *CL_Referenced* variable. Although SEC comment letters occasionally raise issues by explicitly referencing specific Note numbers (e.g., Note 19, as shown in Appendix 1b), this is not always the case. For example, a comment letter may address concerns related to revenue recognition without citing a particular Note number, whereas the CAM report may refer to a specific Note number on that topic. In such cases, both *CL_Revenue* and *CAM_Revenue* would be coded as 1, reflecting alignment at the accounting topic level. However, *CL_Referenced* would remain 0 due to the lack of a shared Note number reference.

²² To clearly distinguish between firms that do not receive CAMs ($CAM=0$) and firms that do receive CAMs ($CAM=1$) but report CAMs unrelated to the topics raised in SEC comment letters, we exclude firms without any CAM reports. For example, a firm may report CAMs solely related to intangible assets ($CAM=1$ and $CAM_Intangible=1$) while receiving an SEC comment letter focused only on revenue recognition ($CL_Revenue=1$). In such cases, we do not interpret this as topic alignment between CAMs and SEC comment letters. Including firms without CAMs ($CAM=0$) would largely reflect non-alignment driven by the absence of CAM reports, which is not the focus of our research design. In other words, our direct link, topic pairwise analysis focuses exclusively on firms reporting at least one CAM to assess whether the topics raised in CAMs correspond to those raised in SEC comment letters.

²³ We are cognizant that endogeneity is endemic to research of this type. To address this concern, we consider the potential effect of correlated omitted variables. Specifically, we test how our main inferences would change if the model were fully specified (e.g., Altonji et al. 2005). Because our main analysis—the association between CAM reports and SEC comment letter issuance—uses a non-linear model, we apply the approach suggested by Cinelli and Hazlett (2020). Our analysis focuses on whether SEC comment letters question the issues referenced in the CAM report (*CL_Referenced*). The results indicate that the robustness value of *CAM* ($RVq=1$) is 0.0424, meaning that unobserved confounders would need to explain more than 4.24% of the residual variance of both CAMs and SEC comment letters. Because Big4 auditor status is one of the key observable control variables in our model, we use it as an example. Specifically, *Big4* has a robustness value of 0.009, which is lower than that of *CAM* (the threshold level), implying that confounders as key as Big4 are not

sufficient to explain away our inferences. Repeating the comparison with other control variables yields similar results (e.g., 0.0159 for *Restatement*, 0.021 for *IC_Weak*, 0.0047 for *HighVolatility* and 0.0008 for SEC Office FE).

²⁴ We measure the EDGAR filing download count on a calendar year basis to ensure temporal comparability across firms with different fiscal year-end dates. Because the SEC's EDGAR access logs are recorded by calendar date, aggregating the data within each calendar year provides a consistent observation window across all firms.

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Appendix 1

a. Example of a CAM in Universal Display Corp.'s 10-K filing for fiscal year 2019

REPORT OF INDEPENDENT REGISTERED PUBLIC ACCOUNTING FIRM

To the Shareholders and Board of Directors
Universal Display Corporation:

Opinion on the Consolidated Financial Statements

We have audited the accompanying consolidated balance sheets of Universal Display Corporation and subsidiaries (the Company) as of December 31, 2019 and 2018, the related consolidated statements of income, comprehensive income, shareholders'

Critical Audit Matter

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that: (1) relates to accounts or disclosures that are material to the consolidated financial statements and (2) involved our especially challenging, subjective, or complex judgment. The communication of a critical audit matter does not affect our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter or on the accounts or disclosures to which it relates.

Notes 2 and 19 regarding the "per unit fee for OLED contracts" are referenced in the CAM report.

Assessment of the estimated per unit fee for long-term OLED contracts

As discussed in Notes 2 and 19 to the consolidated financial statements, the Company recognizes revenue for organic light emitting diode (OLED) sales to customers with long-term contracts (i.e., over 1 year in length) using certain estimates. Revenue is determined by estimating total contract consideration expected to be received over the term of the contract and recognized based on material units sold during the period at their estimated per unit fee. The estimated per unit fee includes fixed amounts designated in contracts with customers as license fees, as well as estimates of material units to be sold and royalties to be earned. The Company uses internal and external data to estimate material units to be sold and royalty consideration to be received over the contract terms.

We identified the assessment of the estimated per unit fee for long-term OLED contracts as a critical audit matter. The estimated per unit fee was dependent upon the estimates of total material units to be sold and royalties to be earned. Significant auditor judgment was required in evaluating the forecasted material unit sales and royalties, as changes in the estimates could significantly affect the estimated per unit fee.

/s/ KPMG LLP

We have served as the Company's auditor since 2002.

Philadelphia, Pennsylvania
February 20, 2020



DIVISION OF
CORPORATION FINANCE

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
WASHINGTON, D.C. 20549

October 19, 2020

Sidney D. Rosenblatt
Chief Financial Officer
Universal Display Corp \PA\
375 Phillips Boulevard
Ewing, NJ 08618

**Re: Universal Display Corp \PA\
Form 10-K for the Year Ended December 31, 2019
Filed February 20, 2020
File No. 001-12031**

SEC comment letter addressing Note 19
regarding the "per unit fee for OLED
contracts"

Dear Mr. Rosenblatt:

We have limited our review of your filing to the financial statements and related disclosures and have the following comments. In some of our comments, we may ask you to

Note 19. Revenue Recognition, page F-34

2. Your revenue recognition policy disclosure starting on page F-13 indicates that material sales are generally recognized at the time the title passes or after control if variable consideration is present. Additionally, technology license agreements and material supply agreements are combined as a single performance obligation and recognized over the contract term based on material units sold at the estimated per unit fee over the life of the contract. Finally, we note your tabular "Revenue Comparison" disclosure in your earnings releases on Form 8-K where you disclose material sales, royalty and license fees, and contract research services. Given it appears you have two material revenue streams, tell us your consideration to disclose disaggregated revenue information in your Forms 10-Q and 10-K pursuant to ASC 606-10-50-5 to 7.

You may contact Effie Simpson at (202) 551-3346 or Melissa Raminpour, Branch Chief at (202) 5513379 with any questions.

Sincerely,

Division of Corporation Finance
Office of Manufacturing



UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
WASHINGTON, D.C. 20549

May 17, 2022

David Lee
Chief Executive Officer
NewHydrogen, Inc.
27936 Lost Canyon Road, Suite 202
Santa Clarita, CA 91387

Re: NewHydrogen, Inc.
Form 10-K for the Fiscal Year Ended December 31, 2021
Filed March 31, 2022
File No. 000-54819

Dear Mr. Lee:

We have limited our review of your filing to the financial statements and related disclosures and have the following comment. In our comment, we may ask you to provide us with information so we may better understand your disclosure.

Please respond to this comment within ten business days by providing the requested information or advise us as soon as possible. Our comment applies to your facts and circumstances.

After reviewing your response to our comment, we have identified the following comment:

SEC Comment letter requiring the firm to disclose how CAMs are addressed in the auditor's report

1. Please amend your Form 10-K to include a revised auditor's report that addresses the following:
 - Ensure the auditors' report is dated as required by AS 3101.10(d) and Rule 2-02(a) of Regulation S-X; and
 - Ensure the auditors' report adequately describes how the critical audit matter was addressed in the audit and refers to the relevant financial statement disclosures that relate to the critical audit matter as required by AS 3101.14(c) and (d). In this regard, it appears stock compensation expense may relate to transactions in addition to stock options.

Sincerely,

Division of Corporation Finance
Office of Manufacturing

Appendix 3

a. Example of a CAM report with a low uncertainty sentiment score.

Critical Audit Matter

The critical audit matter communicated below is a matter that was communicated or required to be communicated to the auditor of the consolidated financial statements and (2) involved our especially difficult, subjective, or complex judgment. The communication of the critical audit matter does not alter in any way our opinion on the consolidated financial statements, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Stating the method used for recognizing revenue

Revenue Recognized Over Time Using Percentage-of-Completion

As described in Note 1 to the consolidated financial statements, the Company generates the majority of its revenue from contracts with customers by units sold with specific specifications and frequencies that are used by a specific customer and contracts where the end user is the United States Government. Control over these performance obligations passes to the customer over time and therefore these revenues are reported in operating results over time using the cost-to-cost method. Under this method, the Company measures progress towards completion and records revenue based on the ratio of costs incurred to date to the estimated total costs.

We identified revenue recognized over time using the percentage-of-completion cost-to cost method as a critical audit matter. Given the judgment necessary to make estimates regarding the revenue and total costs associated with such contracts, specifically, the estimation of costs required to complete such contracts, auditing these estimates required extensive audit effort due to the complexity of the underlying programs and a high degree of auditor judgment when performing audit procedures and evaluating the results of those procedures.

b. Example of CAM report with a high uncertainty sentiment score.

Critical Audit Matters

The critical audit matters communicated below are those that were communicated or required to be communicated to the auditor of the consolidated financial statements and (2) involved our especially difficult, subjective, or complex judgment. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, and we are not, by communicating the critical audit matters below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which they relate.

Discussing the uncertainty arising from (1) the determination of sale or lease; (2) the transaction price; (3) the allocation of transaction price; and (4) the timing of revenue recognition

Revenue Recognition – Theater Systems Revenue

As described in notes 3(p) and 21 to the consolidated financial statements, the Company recognized revenue from IMAX Systems related to the IMAX Technology Sales and Maintenance category (theater systems) of \$65.7 million for the year ended December 31, 2021. Management evaluates whether a theater system arrangement involves either a sale or a lease of a theater system, and for those arrangements that are accounted for as a sale of a theater system, determines the transaction price and the allocation thereof to each separate performance obligation based on estimated standalone selling prices. For arrangements accounted for as a sale of a theater system, the transaction price allocated to the performance obligation is recognized when the conditions signifying transfer of control have been met. For theater system arrangements, management applied significant judgment in (i) determining whether the theater system arrangement related to either a sale or a lease by considering the terms of the arrangement including title to the theater system equipment and payment consideration; (ii) estimating the transaction price which may include the discounted present value of fixed ongoing payments and variable consideration (such as indexed minimum payment increases and additional payments owed by the customer if certain minimum box office receipt thresholds are exceeded); (iii) allocating the transaction price to each separate performance obligation based on estimated standalone selling prices; and (iv) determining the timing of revenue recognition based on when performance obligations are met.

The principal considerations for our determination that performing procedures relating to the revenue recognition of theater systems revenue is a critical audit matter are that management identified the matter as a critical accounting estimate, and there was significant judgment required by management in (i) determining whether the theater system arrangement related to a sale or a lease; (ii) estimating the transaction price which may include the discounted present value of fixed ongoing payments and variable consideration; (iii) allocating the transaction price to each separate performance obligation; and (iv) determining the timing of revenue recognition. This in turn led to a high degree of auditor judgment, subjectivity and effort in performing procedures and evaluating audit evidence relating to the revenue recognition of theater systems revenue.

Appendix 4

Variable Definitions

Variable name	Definition	Data source
CAM	An indicator variable set to 1 if a firm's 10-K filing for year t contains critical audit matters in the auditor's report and 0 otherwise	SEC
CAM_Revenue	An indicator variable set to 1 if a firm's 10-K filing for year t contains critical audit matters related to revenue recognition in the auditor's report and 0 otherwise	SEC
CAM_Intangible	An indicator variable set to 1 if a firm's 10-K filing for year t contains critical audit matters related to intangible assets in the auditor's report and 0 otherwise	SEC
CAM_M&A	An indicator variable set to 1 if a firm's 10-K filing for year t contains critical audit matters related to mergers and acquisitions or business combinations in the auditor's report and 0 otherwise	SEC
Uncertainty_Revenue	The uncertainty sentiment score of CAM reports pertaining to revenue recognition. The score is based on RoBERTa architecture, a language representation model developed by Meta. For a detailed description of the model, see https://huggingface.co/docs/transformers/en/model_doc/roberta .	SEC
Uncertainty_Intangible	The uncertainty sentiment score of CAM reports pertaining to intangible assets. The score is based on RoBERTa architecture, a language representation model developed by Meta.	SEC
Uncertainty_M&A	The uncertainty sentiment score of CAM reports pertaining to M&A. The score is based on RoBERTa architecture, a language representation model developed by Meta.	SEC
CL_Referenced	An indicator variable set to 1 if a firm receives an SEC comment letter raising issues referenced in the auditor's CAM report for year t and 0 otherwise	Audit Analytics
CL_Supervisor	An indicator variable set to 1 if a firm's comment letter process involves a supervisor and 0 otherwise	SEC
CL_Revenue	An indicator variable set to 1 if a firm receives an SEC comment letter raising issues related to revenue recognition for year t 's 10-K filing and 0 otherwise	Audit Analytics
CL_Intangible	An indicator variable set to 1 if a firm receives an SEC comment letter raising issues related to intangible assets for year t 's 10-K filing and 0 otherwise	Audit Analytics
CL_M&A	An indicator variable set to 1 if a firm receives an SEC comment letter raising issues related to mergers and acquisitions or business combinations for year t 's 10-K filing and 0 otherwise	Audit Analytics
2Tier	An indicator variable set to 1 if a firm is audited by a second-tier audit firm (BDO, Crowe Horwath, Grant Thornton or McGladrey) and 0 otherwise	Audit Analytics
Age	The natural logarithm of firm age in years relative to the first year the firm appears in Compustat	Compustat
Big4	An indicator variable set to 1 if a firm is audited by a Big 4 auditor in year t and 0 otherwise	Audit Analytics
Chg_Sales	Percentage change in sales (REVT) between year $t + 1$ to year t	Compustat
External_Financing	The sum of equity financing and debt financing scaled by total assets in year $t + 1$	Compustat
HighVolatility	An indicator variable set to 1 if the volatility of abnormal monthly stock returns (i.e., monthly returns [RET] minus the value-weighted returns [VWRTD]) is in the highest quartile for the fiscal year and 0 otherwise. We measure volatility as the standard deviation of monthly stock returns over a 12-month window ending the last month of the fiscal year.	CRSP

Variable name	Definition	Data source
IC_Weak	An indicator variable set to 1 if the internal control audit opinion (under SOX Section 404) or the management certification (under SOX Section 302) is qualified because of a material weakness in year t and 0 otherwise	Audit Analytics
Loss	An indicator variable set to 1 when earnings before extraordinary items (IB) in year t are negative and 0 otherwise	Compustat
M&A	An indicator variable set to 1 if a firm undertakes acquisitions or mergers (AQP) in year t and 0 otherwise	Compustat
Restatement	An indicator variable set to 1 if a firm restates its 10-K in year t and 0 otherwise	Audit Analytics
Restructure	An indicator variable set to 1 if a firm reports restructuring costs (RCP) in year t and 0 otherwise	Compustat
SEC_Download	The natural logarithm of the number of a firm's Form 10-K downloads from EDGAR (SEC.gov) in a calendar year. EDGAR log file data are available at https://www.sec.gov/data-research/sec-markets-data/edgar-log-file-data-sets .	SEC
Segment	The natural logarithm of the number of business and geographic segments	Compustat
Size	The natural logarithm of market capitalization, calculated as shares outstanding at fiscal year-end (CSHO) times the share price at fiscal year-end (PRCC_F)	Compustat
ZScore	Decile rank of the firm's Altman's Z-score for the year. Firms with the poorest (highest) financial health have a value of 10 (1). We calculate the Altman's Z-score (Altman 1968) as $1.2 * [\text{net working capital (ACT-LCT)/total assets (AT)}] + 1.4 * [\text{retained earnings (RE)/total assets (AT)}] + 3.3 * [\text{earnings before interest and taxes (PI+XINT)/total assets (AT)}] + 0.6 * [\text{market value of equity (CSHO * PRCC_F)/book value of liabilities (LT)}] + 1.0 * [\text{sales (SALE)/total assets (AT)}]$.	Compustat